Descriptor-Teacher Guided Multi-to-Uni Transfer for Low-Resource Regression ADMET Prediction

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Abstract

Low-resource ADMET prediction is a recurring challenge in early-stage drug discovery, where labeled assays are limited but inexpensive molecular knowledge such as physicochemical descriptors is readily available. We study whether descriptor knowledge can be transferred into an ECFP-only neural predictor, so that descriptors guide training without being required at inference time. We develop Descriptor-Teacher AdapterFusion, a lightweight ECFP4-based student model that combines an adapter-based pseudo-descriptor representation with descriptor reconstruction and teacher prediction distillation from an ECFP4+descriptor random forest. Across five regression ADMET datasets, three low-resource training ratios, and three random seeds, fixed teacher distillation with $\lambda_{\text{distill}} = 1.0$ improves AdapterFusion in 11/15 dataset-ratio settings, with an average RMSE change of -0.0656 relative to the undistilled AdapterFusion baseline. Descriptor-access random forest models remain stronger in most settings, so the contribution is controlled descriptor knowledge transfer rather than an overall ADMET state-of-the-art claim. A lambda sweep and adaptive-weighting ablation further show that simple validation-based adaptive distillation is not sufficient.

1 Introduction

Accurate ADMET prediction is important for early drug discovery because absorption, distribution, metabolism, excretion, and toxicity properties affect which candidate molecules can progress beyond screening [2, 18, 20]. Many ADMET tasks are low-resource: labels are expensive to obtain, measurements vary across assays, and each property may have a limited number of reliable annotated compounds. This makes it useful to study lightweight models and auxiliary molecular knowledge that can improve generalization under limited supervision.

Molecular descriptors provide one such source of auxiliary knowledge. RDKit physicochemical descriptors are inexpensive to compute and often correlate with ADMET behavior [13]. A direct way to exploit them is to concatenate descriptors with molecular fingerprints or use descriptor-based traditional models. However, this creates a descriptor-access inference setting. In contrast, a multi-to-uni transfer setting asks whether descriptor knowledge can be used during training while the deployed model uses only a single modality, here ECFP4 fingerprints [14].

This paper studies descriptor-guided multi-to-uni transfer for low-resource ADMET prediction. We begin from an ECFP4 MLP baseline, compare simple auxiliary descriptor prediction with a structured AdapterFusion student, and then add teacher prediction distillation from an ECFP4+descriptor random forest. Our central question is deliberately narrow: can descriptor-access knowledge improve an ECFP-only student in low-resource regression ADMET? The answer is positive but bounded. Teacher distillation improves AdapterFusion in most regression settings, but descriptor-access random forest baselines remain strong.

The contributions are:

- We formulate a lightweight descriptor-guided multi-to-uni transfer setup for low-resource ADMET prediction.
- We compare ECFP-only descriptor prediction, AdapterFusion, descriptor-access controls, random forest baselines, and teacher-distilled AdapterFusion under shared low-resource splits.
- We show that fixed descriptor-teacher distillation improves regression AdapterFusion in 11/15 dataset-ratio settings.
- We report lambda sensitivity, validation-selected lambda analysis, and a negative adaptive-weighting ablation.

2 Related Work

Molecular representation learning has increasingly used auxiliary objectives to inject chemical knowledge into learned representations. Molecular language models such as MolBERT motivate the use of domain-relevant pretraining or auxiliary tasks rather than relying only on generic representation learning objectives [5]. Graph pretraining methods such as GNN pretraining, GROVER, and GraphMVP similarly use self-supervised or cross-view objectives to improve downstream molecular prediction [8, 12, 15]. Our work is smaller in scale and does not use a large pretrained encoder, but it shares the principle that chemically meaningful auxiliary targets can shape molecular representations.

Auxiliary learning and task-specific adaptation are also relevant. A simple auxiliary objective may not transfer all useful knowledge to the downstream task; the architecture through which auxiliary information is routed can matter [4, 16]. The closest conceptual motivation is multi-to-uni molecular representation learning, including M2UMol-style knowledge transfer [21]. Our setting is deliberately controlled: descriptors are the auxiliary modality, ECFP4 is the inference modality, and ADMET prediction is the downstream task.

ADMET benchmark modeling provides the application context. We use TDC-style ADMET datasets and evaluate low-resource splits across regression and classification properties [9]. MoleculeNet provides an earlier benchmarking reference for molecular property prediction metrics, splits, and baselines [19]. Fingerprint and descriptor models remain important controls in cheminformatics, while graph neural networks and directed message-passing models provide stronger learned-representation baselines in larger studies [6, 11, 22, 23]. Tree ensembles are included because random forests and gradient boosting methods remain competitive for tabular molecular features [1, 3, 10, 17].

3 Method

3.1 Problem setting

Each molecule has an ECFP4 fingerprint x_{ecfp}, an optional descriptor vector x_{desc}, and a target y. The primary inference constraint is that the student model should use only x_{ecfp} at test time. Descriptor information may be used during training as auxiliary supervision or through teacher predictions. The main paper focuses on regression tasks, where the task loss is mean squared error.

3.2 ECFP encoder and descriptor prediction

The base ECFP model maps a 2048-dimensional ECFP4 fingerprint to a 128-dimensional latent representation using a two-layer MLP:

$$z_{\text{fp}} = f_{\text{fp}}(x_{\text{ecfp}}). \tag{1}$$

The simple descriptor-prediction baseline adds an auxiliary head that predicts standardized RDKit descriptors from z_{fp}. Its training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{transfer}}\mathcal{L}_{\text{desc}}, \tag{2}$$

where $\lambda_{\text{transfer}} = 0.1$. The descriptor vector contains nine RDKit descriptors: MolWt, LogP, TPSA, HBA, HBD, rotatable bonds, aromatic ring count, heavy atom count, and ring count.

3.3 AdapterFusion student

The AdapterFusion student uses the same ECFP encoder but introduces a pseudo-descriptor adapter:

$$z_{\text{desc}} = f_{\text{adapter}}(z_{\text{fp}}). \tag{3}$$

A learned fusion gate combines the original ECFP representation and the pseudo-descriptor representation:

$$g = \sigma(f_{\text{gate}}([z_{\text{fp}}, z_{\text{desc}}])), \tag{4}$$

$$z_{\text{fused}} = g \odot z_{\text{fp}} + (1 - g) \odot z_{\text{desc}}. \tag{5}$$

The target prediction head receives z_{fused}, while the descriptor prediction head predicts descriptors from z_{desc}. This structure keeps inference ECFP-only, because no true descriptors are consumed by the student at test time.

3.4 Descriptor-teacher distillation

We include descriptor-access controls to estimate the value of direct descriptor information. The strongest descriptor-access teacher used here is an ECFP4+descriptor random forest. The distilled student keeps the AdapterFusion architecture and adds a teacher prediction loss:

$$\mathcal{L}_{\text{teacher}} = \text{MSE}(\hat{y}_{\text{student}}, \hat{y}_{\text{teacher}}). \tag{6}$$

The final objective is:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{transfer}}\mathcal{L}_{\text{desc}} + \lambda_{\text{distill}}\mathcal{L}_{\text{teacher}}. \tag{7}$$

The main configuration uses $\lambda_{\text{transfer}} = 0.1$ and $\lambda_{\text{distill}} = 1.0$, selected from the fixed lambda sweep over $\{0.01, 0.1, 0.3, 1.0\}$. This follows the broader teacher-student idea of knowledge distillation [7].

4 Experiments

4.1 Datasets and protocol

The main regression benchmark includes five TDC ADMET tasks: Caco-2 permeability, lipophilicity, aqueous solubility, volume of distribution, and plasma protein binding. Supplementary classification experiments cover five standard binary ADMET endpoints. The main claims are restricted to regression tasks because the distillation signal is more consistent there.

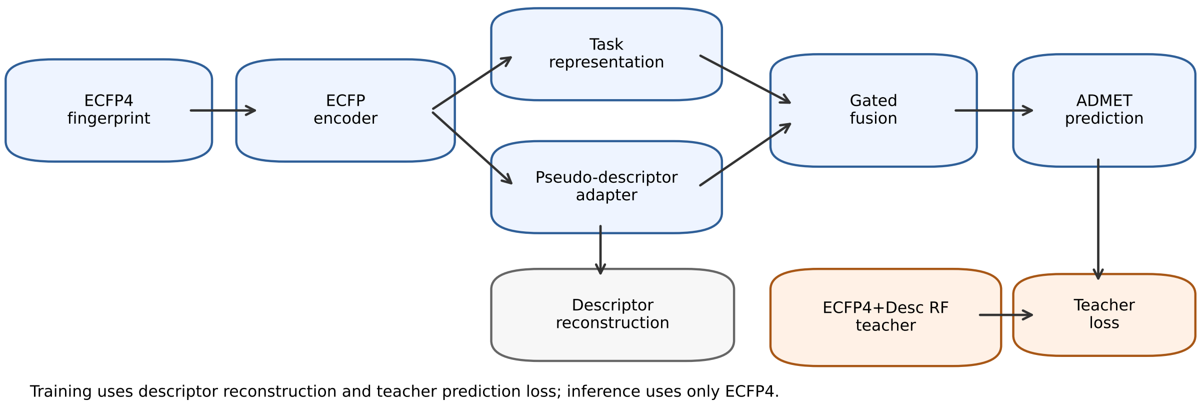

Figure 1: Descriptor-Teacher AdapterFusion. Training uses descriptor reconstruction and teacher prediction distillation from an ECFP4+descriptor random forest. Inference uses only ECFP4.

Each dataset is evaluated with training ratios of 10%, 20%, and 50%. Each setting is run with seeds 42, 123, and 3407. All models use the same dataset-ratio-seed splits. Descriptor scalers are fit only on the training split and then applied to validation and test splits.

4.2 Baselines and metrics

We compare ECFP4 MLP, ECFP4 MLP with descriptor prediction, AdapterFusion, descriptor-concatenation MLP, ECFP4 random forest, descriptor random forest, ECFP4+descriptor random forest, and Descriptor-Teacher AdapterFusion. Descriptor-concatenation and descriptor random forest models use descriptors at inference and should be interpreted as descriptor-access controls rather than ECFP-only students.

Regression tasks use MAE and RMSE, with RMSE as the primary metric. Classification tasks use ROC-AUC and PR-AUC and are reported only as supplementary reference results. All tables report mean±standard deviation over three seeds.

5 Results

5.1 Main regression results

Table 1 shows the main low-resource regression results. Descriptor-access random forests remain strong, confirming that descriptor information is predictive for ADMET properties. Among ECFP-only neural models, AdapterFusion is consistently stronger than simple descriptor prediction on the regression tasks, supporting the hypothesis that descriptor knowledge is better transferred through a structured pseudo-descriptor adapter than through a plain auxiliary descriptor head.

Teacher distillation further improves AdapterFusion. With $\lambda_{\text{distill}} = 1.0$, distilled AdapterFusion improves over base AdapterFusion in 11/15 regression settings, with a mean RMSE delta of -0.0656. The gains are most visible in settings where the descriptor-access teacher has a large advantage over the ECFP-only student, such as `ppbr_az`. The distilled student remains below the descriptor-access

Table 1: Main low-resource regression ADMET results. Values are test RMSE mean±std over three seeds; lower is better. Δ is distilled AdapterFusion minus base AdapterFusion, so negative values indicate improvement.

Dataset	Train %	AdapterFusion	ECFP4+Desc RF	Distilled AdapterFusion	Δ
caco2_wang	10	1.7702±0.0304	0.5535±0.0017	1.7725±0.0497	0.0023
caco2_wang	20	0.9541±0.0563	0.4691±0.0083	0.9459±0.0362	-0.0081
caco2_wang	50	0.6775±0.0995	0.4252±0.0121	0.6265±0.0508	-0.0510
lipophilicity_astrazeneca	10	1.2325±0.0114	1.0635±0.0011	1.2058±0.0219	-0.0267
lipophilicity_astrazeneca	20	1.0960±0.0132	0.9858±0.0015	1.0903±0.0083	-0.0057
lipophilicity_astrazeneca	50	0.9736±0.0059	0.9184±0.0012	0.9605±0.0057	-0.0130
solubility_aqsoldb	10	1.7832±0.0151	1.4996±0.0059	1.7906±0.0205	0.0074
solubility_aqsoldb	20	1.7117±0.0125	1.3951±0.0021	1.7027±0.0288	-0.0090
solubility_aqsoldb	50	1.6107±0.0088	1.3397±0.0057	1.5963±0.0087	-0.0144
vdss_lombardo	10	5.4563±0.0463	5.2699±0.0051	5.4187±0.0199	-0.0376
vdss_lombardo	20	5.5086±0.1260	5.2805±0.0220	5.6795±0.0473	0.1709
vdss_lombardo	50	5.2838±0.0352	5.0729±0.0257	5.3103±0.0309	0.0265
ppbr_az	10	18.1281±0.1596	14.2419±0.0867	17.5888±0.2565	-0.5393
ppbr_az	20	16.6702±0.3270	13.7317±0.0288	16.4194±1.1199	-0.2508
ppbr_az	50	14.9058±0.5019	13.3106±0.0968	14.6711±0.7282	-0.2347

Table 2: Fixed distillation lambda ablation over 15 regression dataset-ratio settings. Negative mean delta indicates improvement over base AdapterFusion.

$\lambda_{distill}$	Complete	Beats Adapter	Mean Δ
0.01	15/15	8/15	0.0075
0.1	15/15	8/15	-0.0221
0.3	15/15	10/15	-0.0046
1.0	15/15	11/15	-0.0656

RF teacher in most settings, so the result should be interpreted as partial descriptor knowledge transfer, not replacement of descriptor-access baselines.

5.2 Lambda sensitivity

Table 2 summarizes the fixed distillation lambda sweep. The strongest global fixed value is $\lambda_{\text{distill}} = 1.0$. Smaller lambdas can be better for individual settings, but they do not improve aggregate performance. Validation-selected lambda is useful as an analysis but beats fixed $\lambda_{\text{distill}} = 1.0$ in only 2/15 settings, while still beating base AdapterFusion in 12/15 settings.

5.3 Adaptive weighting as a negative ablation

Table 3 summarizes the validation teacher-advantage adaptive weighting rule. The adaptive rule improves over base AdapterFusion in 9/15 settings, but beats fixed $\lambda_{\text{distill}} = 1.0$ in only 3/15 settings and beats the best fixed lambda in only 1/15 settings. Its mean RMSE delta relative to fixed $\lambda_{\text{distill}} = 1.0$ is 0.0817, indicating worse average performance. This negative result shows that raw validation teacher advantage is not a sufficient reliability signal for adaptive distillation.

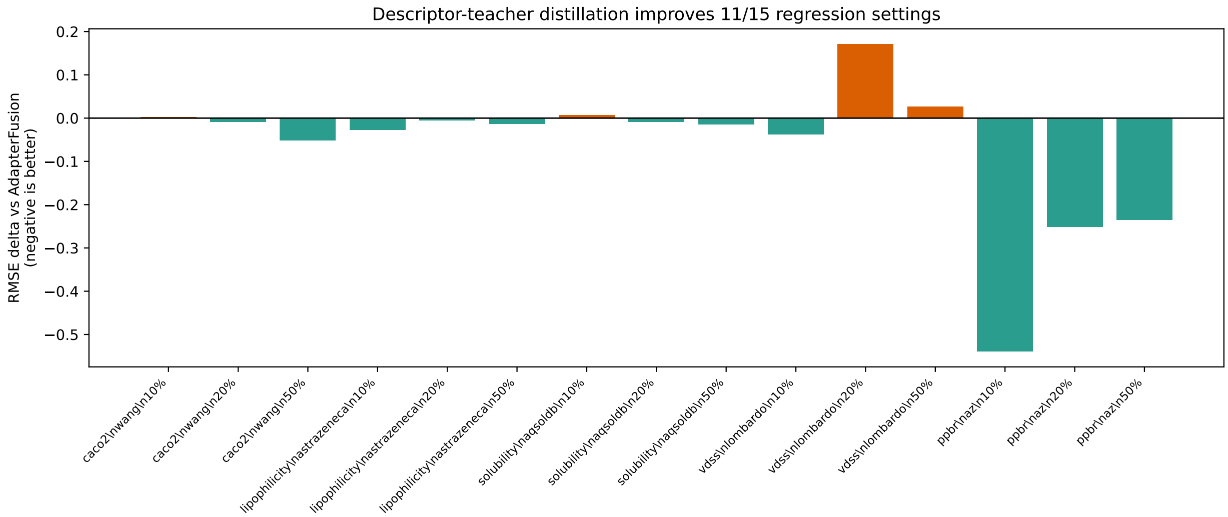

Figure 2: RMSE delta of distilled AdapterFusion relative to base AdapterFusion. Negative values indicate improvement. Fixed $\lambda_{\text{distill}} = 1.0$ improves 11/15 regression settings.

Table 3: Adaptive teacher-weighting negative ablation. The validation-advantage rule improves over base AdapterFusion in some settings, but is weaker than fixed strong distillation.

Comparison	Result
Adaptive beats base AdapterFusion	9/15
Adaptive beats fixed $\lambda = 1.0$	3/15
Adaptive beats best fixed lambda	1/15
Mean Δ vs fixed $\lambda = 1.0$	0.0817

5.4 Classification reference

Classification results are reported as supplementary results. They are less stable than the regression results, and descriptor-access controls are often strong. We therefore do not use classification results as the main evidence for Descriptor-Teacher AdapterFusion.

6 Discussion

The experiments support a controlled descriptor-guided transfer story. Descriptor information is valuable, and descriptor-access baselines remain strong. The contribution is not to replace descriptor-access random forests, but to show that descriptor-access knowledge can partially improve an ECFP-only neural student under low-resource regression ADMET settings.

AdapterFusion improves over plain descriptor prediction, suggesting that the structure of auxiliary transfer matters. Teacher distillation then provides an additional improvement by aligning the ECFP-only student with predictions from a descriptor-access random forest teacher. The lambda sweep shows that stronger teacher supervision is generally useful, with $\lambda_{\text{distill}} = 1.0$ as the best fixed global setting. However, the dataset-specific best lambda varies, and neither validation-selected lambda nor the tested adaptive rule clearly improves over the fixed setting.

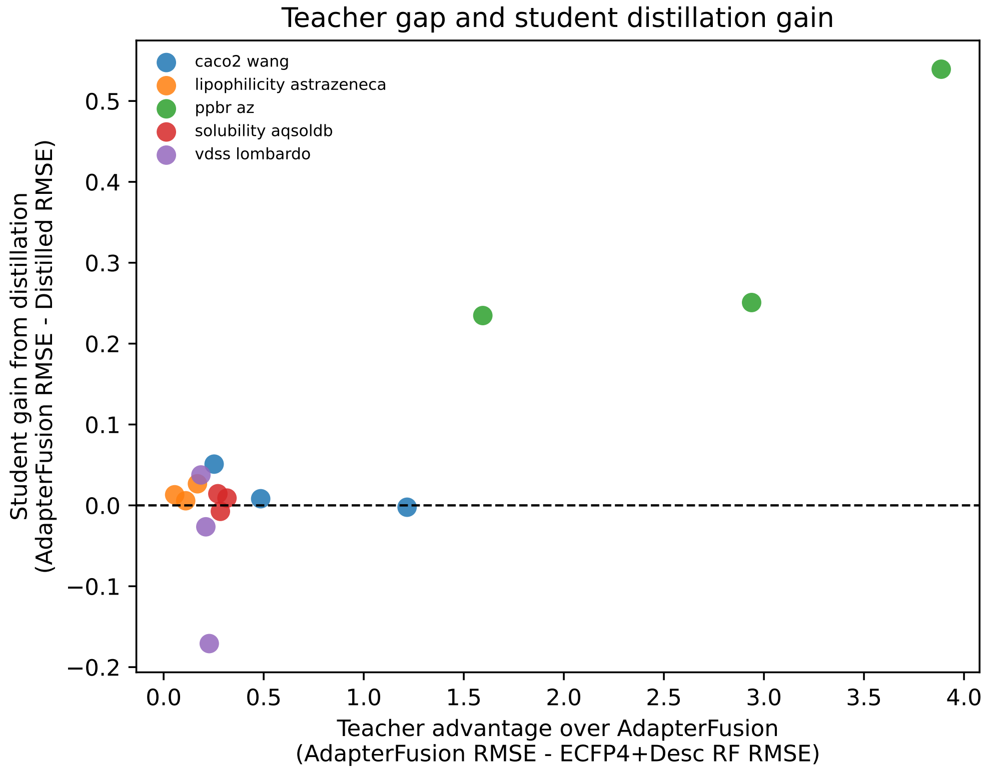

Figure 3: Relationship between descriptor-teacher advantage and student gain from distillation. Positive student gain means the distilled student has lower RMSE than base AdapterFusion.

7 Limitations

This work has several limitations. First, the main claim is restricted to regression ADMET tasks. Classification results are included only as supplementary evidence because improvements are less consistent. Second, the student model is lightweight and ECFP-based; the method has not yet been tested with large pretrained molecular encoders. Third, descriptor-access random forests remain stronger than the distilled student in most settings. Finally, the adaptive teacher-weighting strategy tested here is simple and negative; more principled sample-level or learned teacher weighting may be needed.

8 Conclusion

We presented Descriptor-Teacher AdapterFusion, a lightweight descriptor-guided multi-to-uni transfer method for low-resource regression ADMET prediction. The method trains an ECFP-only AdapterFusion student with descriptor reconstruction and teacher prediction distillation from an ECFP4+descriptor random forest. The main fixed setting, $\lambda_{\text{distill}} = 1.0$, improves AdapterFusion in 11/15 regression dataset-ratio settings. At the same time, descriptor-access random forest baselines remain stronger, so the appropriate conclusion is controlled descriptor knowledge transfer rather than overall ADMET state-of-the-art performance.

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